2.53 Forces: Coefficient of Friction

Worked Solutions

1 a)

$R(\uparrow)$: $R - 50 = 0 \Rightarrow R = 50$

$F = 0.2 \times 50 = 10$

1 b)

$R(\uparrow)$: $R - 20 = 0 \Rightarrow R = 20$

$F = 0.45 \times 20 = 9$

1 c)

$R(\uparrow)$: $R - 15 = 0 \Rightarrow R = 15$

$F = 0.85 \times 15 = 12.75$

1 d)

$R(\uparrow)$: $R - 3 = 0 \Rightarrow R = 3$

$F = 1.1 \times 3 = 3.3$

2 a)

$R(\uparrow)$: $R - 12 = 0 \Rightarrow R = 12$

$F = 0.5 \times 12 = 6$

$R(\rightarrow)$: $P - 6 = 0 \Rightarrow P = 6$

2 b)

$R(\uparrow)$: $R - 12 = 0 \Rightarrow R = 12$

$F = 0.9 \times 12 = 10.8$

$R(\rightarrow)$: $P - 10.8 = 0 \Rightarrow P = 10.8$

2 c)

$R(\rightarrow)$: $20 - F = 0 \Rightarrow F = 20$

$20 = 0.4 \times R \Rightarrow R = 50$

$R(\uparrow)$: $50 - w = 0 \Rightarrow w = 50$

2 d)

$R(\rightarrow)$: $20 - F = 0 \Rightarrow F = 20$

$20 = 0.8 \times R \Rightarrow R = 25$

$R(\uparrow)$: $25 - w = 0 \Rightarrow w = 25$

2 e)

$R(\uparrow)$: $R - 50 = 0 \Rightarrow R = 50$

$R(\rightarrow)$: $40 - F = 0 \Rightarrow F = 40$

$40 = \mu \times 50 \Rightarrow \mu = 0.8$

2 f)

$R(\uparrow)$: $R - 300 = 0 \Rightarrow R = 300$

$R(\rightarrow)$: $12 - F = 0 \Rightarrow F = 12$

$12 = \mu \times 300 \Rightarrow \mu = 0.04$

2 g)

$R(\uparrow)$: $R - 12 = 0 \Rightarrow R = 12$

$R(\rightarrow)$: $15 - F = 0 \Rightarrow F = 15$

$15 = \mu \times 12 \Rightarrow \mu = 1.25$

2 h)

$R(\uparrow)$: $R - 12 - 3 = 0 \Rightarrow R = 15$

$F = 0.7 \times 15 = 10.5$

$R(\rightarrow)$: $P - 10.5 = 0 \Rightarrow P = 10.5$

2 i)

$R(\rightarrow)$: $8 - F = 0 \Rightarrow F = 8$

$8 = 0.8 \times R \Rightarrow R = 10$

$R(\uparrow)$: $10 - 2 - w = 0 \Rightarrow w = 8$

2 j)

$R(\uparrow)$: $R - 20 - 60 = 0 \Rightarrow R = 80$

$R(\rightarrow)$: $10 - F = 0 \Rightarrow F = 10$

$10 = \mu \times 80 \Rightarrow \mu = 0.125$

3 a)

$R(\uparrow)$: $R + 15\sin 30^\circ - 20 = 0 \Rightarrow R = 12.5$

$F = 0.5 \times 12.5 = 6.25$

3 b)

$R(\uparrow)$: $R + T\sin 45^\circ - 8 = 0$

$F = 0.2 \times (8 - \frac{\sqrt{2}}{2}T) = \frac{8}{5} - \frac{\sqrt{2}}{10}T$

$R(\rightarrow)$: $T\cos 45^\circ - \frac{8}{5} + \frac{\sqrt{2}}{10}T = 0$

$\frac{\sqrt{2}}{2}T - \frac{8}{5} + \frac{\sqrt{2}}{10}T = 0$

$\Rightarrow T = \frac{4\sqrt{2}}{3} = 1.89$ (3 SF)

3 c)

$F = 0.1R$

$R(\rightarrow)$: $10\cos 60^\circ - 0.1R = 0 \Rightarrow R = 50$

$R(\uparrow)$: $50 + 10\sin 60^\circ - w = 0$

$\Rightarrow w = 50 + 5\sqrt{3} = 58.7$ (3 SF)

3 d)

$R(\uparrow)$: $R + 4\sin 30° - 15 = 0 \Rightarrow R = 13$

$R(\rightarrow)$: $4\cos 30° - F = 0 \Rightarrow F = 2\sqrt{3}$

$2\sqrt{3} = \mu \times 13 \Rightarrow \mu = \frac{2\sqrt{3}}{13} = 0.266$ (3 SF)

4 a)

$R(\uparrow)$: $R - 4\sin 30° - 2 = 0 \Rightarrow R = 4$

$F = \frac{1}{2}\sqrt{3} \times 4 = 2\sqrt{3} = 3.46$ (3 SF)

4 b)

$R(\uparrow)$: $R - T\sin 45° - 4 = 0$

$F = 0.3 \times (4 + \frac{\sqrt{2}}{2}T) = \frac{6}{5} + \frac{3\sqrt{2}}{20}T$

$R(\rightarrow)$: $T\cos 45° - \frac{6}{5} - \frac{3\sqrt{2}}{20}T = 0$

$\frac{\sqrt{2}}{2}T - \frac{6}{5} - \frac{3\sqrt{2}}{20}T = 0$

$\Rightarrow T = \frac{12\sqrt{2}}{7} = 2.42$ (3 SF)

4 c)

$F = 0.2R$

$R(\rightarrow)$: $12\cos 60° - 0.2R = 0 \Rightarrow R = 30$

$R(\uparrow)$: $30 - 12\sin 60° - w = 0$

$\Rightarrow w = 30 - 6\sqrt{3} = 19.6$ (3 SF)

4 d)

$R(\uparrow)$: $R - 5\sin 30° - 20 = 0 \Rightarrow R = 22.5$

$R(\rightarrow)$: $5\cos 30° - F = 0 \Rightarrow F = \frac{5\sqrt{3}}{2}$

$\frac{5\sqrt{3}}{2} = \mu \times 22.5 \Rightarrow \mu = \frac{\sqrt{3}}{9} = 0.192$ (3 SF)

5 a)

$R(\uparrow)$: $25 + 14\sin 27° - w = 0 \Rightarrow w = 31.4$

This is the weight and not the mass of the particle.

6 a)

$R(\uparrow)$: $R + T\sin 10° - 5 \times 9.8 = 0$

$F = 0.5 \times (49 - T\sin 10°)$

$R(\rightarrow)$: $T\cos 10° - 0.5(49 - T\sin 10°) = 0$

$T\cos 10° - 24.5 + 0.5T\sin 10° = 0$

$(\cos 10° + 0.5\sin 10°)T = 24.5$

$\Rightarrow T = 22.9$ (3 SF)

$T = 23$ (2 sf – AQA)

6 b)

$F = 0.1R$

$R(\rightarrow)$: $20\cos 50° - 0.1R = 0 \Rightarrow R = 200\cos 50°$

$R(\uparrow)$: $200\cos 50° + 20\sin 50° - m \times 9.8 = 0$

$\Rightarrow m = 14.7$ (3 SF)

$m = 15$ (2 sf – AQA)

6 c)

$R(\uparrow)$: $R + 5\sin 70° - 2 \times 9.8 = 0$

$\Rightarrow R = 19.6 - 5\sin 70°$

$R(\rightarrow)$: $5\cos 70° - F = 0 \Rightarrow F = 5\cos 70°$

$5\cos 70° = \mu \times (19.6 - 5\sin 70°)$

$\Rightarrow \mu = 0.115$ (3 SF)

$\mu = 0.11$ (2 sf – AQA)

7 a)

$R(\uparrow)$: $R - 2 \times 9.8 = 0 \Rightarrow R = 19.6$

$F = 0.4 \times 19.6 = 7.84$

$R(\rightarrow)$: $8.4 - 7.84 = 0.56 > 0$ so the particle moves.

$R(\uparrow)$: $R - 0.1 \times 9.8 = 0 \Rightarrow R = 0.98$

$F = 0.4 \times 0.98 = 0.392$

$R(\rightarrow)$: $8.4 - 7.84 - 0.392 = 0.168 > 0$ so the particle continues to move.

7 b)

$R(\uparrow)$: $R - 2 \times 9.8 = 0 \Rightarrow R = 19.6$

$F = 0.4 \times 19.6 = 7.84$

$R(\rightarrow)$: $8 - 7.84 = 0.16 > 0$ so the particle moves.

$R(\uparrow)$: $R - 0.1 \times 9.8 = 0 \Rightarrow R = 0.98$

$F = 0.4 \times 0.98 = 0.392$

$R(\rightarrow)$: $8 - 7.84 - 0.392 = -0.232 < 0$ so the particle no longer moves.

7 c)

$R(\uparrow)$: $R - 2 \times 9.8 = 0 \Rightarrow R = 19.6$

$F = 0.4 \times 19.6 = 7.84$

$R(\rightarrow)$: $7.6 - 7.84 = -0.24 < 0$ so the particle doesn't move.

Adding more mass, means it continues to not move.

8 a)

$R(\uparrow)$: $R - 5 \times 9.8 = 0 \Rightarrow R = 49$

$F = 0.3 \times 49 = 14.7$

$R(\rightarrow)$: $P - 14.7 = 0 \therefore P_{\max} = 14.7$

$P_{\max} = 15$ (2 SF – AQA)

8 b)

$R(\uparrow)$: $R - 10 \times 9.8 = 0 \Rightarrow R = 98$

$F = 0.3 \times 98 = 29.4$

$R(\rightarrow)$: $P - 29.4 = 0 \therefore P_{\max} = 29.4$

$P_{\max} = 29$ (2 SF – AQA)

8 c)

$R(\uparrow)$: $R - 20 \times 9.8 = 0 \Rightarrow R = 196$

$F = 0.3 \times 196 = 58.8$

$R(\rightarrow)$: $P - 58.8 = 0 \therefore P_{\max} = 58.8$

$P_{\max} = 60$ (2 SF – AQA)

9)

$R(\uparrow)$: $R - mg = 0 \Rightarrow R = mg$

$F = \mu \times mg$

$R(\rightarrow)$: $P - \mu mg = 0$ (at point of moving)

$P - \mu mg \leq 0$ so $P \leq \mu mg$ as req.

10)

$R(\uparrow)$: $R + 10\sin\theta - 150 = 0$

$\Rightarrow R = 150 - 10\sin\theta$

$F = 0.1 \times (150 - 10\sin\theta) = 15 - \sin\theta$

$R(\rightarrow)$: $10\cos\theta - 15 + \sin\theta > 0$

$10\cos\theta + \sin\theta > 15$ as req

11 a)

$R(\uparrow)$: $R - 15 \times 10 = 0 \Rightarrow R = 150$

$F = 0.85 \times 150 = 127.5$

$R(\rightarrow)$: $120 - 127.5 < 0 \therefore$ does not move

11 b)

$R(\uparrow)$: $R + 120\sin 30° - 15 \times 10 = 0 \Rightarrow R = 90$

$F = 0.85 \times 90 = 76.5$

$R(\rightarrow)$: $120\cos 30° - 76.5 = 15a$

$\Rightarrow a = 1.83$ ms^{-2}

$a = 2$ ms^{-2} (1 SF – AQA)

11 c)

$R(\uparrow)$: $R + 120\sin 30° - 15 \times 10 - 10m = 0$

$\Rightarrow R = 90 + 10m$

$F = 0.85 \times (90 + 10m) = 76.5 + 8.5m$

$R(\rightarrow)$: $120\cos 30° - 76.5 - 8.5m > 0$

$\Rightarrow 120\cos 30° - 76.5 > 8.5m$

$m < 3.226240995\ldots$

$m < 3.23$

$m < 3$ (1 SF – AQA)

12)

$R(\uparrow)$: $R + T\sin\theta - mg = 0 \Rightarrow R = mg - T\sin\theta$

$F = \mu(mg - T\sin\theta)$

$R(\rightarrow)$: $T\cos\theta - \mu mg + \mu T\sin\theta = 0$

$T\cos\theta + \mu T\sin\theta = \mu mg$

$T(\cos\theta + \mu\sin\theta) = \mu mg$

$T = \frac{\mu mg}{\cos\theta + \mu\sin\theta}$

Therefore, for the box to be moving

$T \geq \frac{\mu mg}{\cos\theta + \mu\sin\theta}$ as req.

13 a)

$\sqrt{3^2 + 4^2} = 5$ so $\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$

13 b)

$\sqrt{5^2 + 12^2} = 13$ so $\sin\theta = \frac{5}{13}, \cos\theta = \frac{12}{13}$

13 c)

$\sqrt{7^2 + 16^2} = \sqrt{305}$ so $\sin\theta = \frac{7}{\sqrt{305}}, \cos\theta = \frac{16}{\sqrt{305}}$

13 d)

$\sqrt{10^2 + 21^2} = \sqrt{541}$ so $\sin\theta = \frac{10}{\sqrt{541}}, \cos\theta = \frac{21}{\sqrt{541}}$

14 a)

$\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$

$R(\uparrow)$: $R + 20 \times \frac{3}{5} - 10 \times 9.8 = 0 \Rightarrow R = 86$

$F = 0.1 \times 86 = 8.6$

$R(\rightarrow)$: $20 \times \frac{4}{5} - 8.6 = 10a \Rightarrow a = 0.74$

14 b)

$\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$

$R(\uparrow)$: $R + 20 \times \frac{3}{5} - 3 \times 9.8 = 0 \Rightarrow R = 17.4$

$F = 0.2 \times 17.4 = 3.48$

$R(\rightarrow)$: $20 \times \frac{4}{5} - 3.48 = 3a \Rightarrow a = \frac{313}{75} = 4.17$ (3 SF)

$a = 4.2$ (2 SF – AQA)

14 c)

$\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$

$R(\uparrow)$: $R + 15 \times \frac{3}{5} - 5 \times 9.8 = 0 \Rightarrow R = 40$

$R(\rightarrow)$: $15 \times \frac{4}{5} - F = 5 \times 1.5 \Rightarrow F = 4.5$

$4.5 = \mu \times 40 \Rightarrow \mu = 0.1125$

$\mu = 0.11$ (2 SF – AQA)

14 d)

$\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$

$R(\rightarrow)$: $16 \times \frac{4}{5} - F = 3m \Rightarrow F = 12.8 - 3m$

$R(\uparrow)$: $R + 16 \times \frac{3}{5} - 9.8m = 0 \Rightarrow R = 9.8m - 9.6$

$12.8 - 3m = 0.25 \times (9.8m - 9.6)$

$12.8 - 3m = 2.45m - 2.4$

$\Rightarrow m = \frac{304}{109} = 2.79$ (3 SF)

$m = 2.8$ (2 SF – AQA)

14 e)

$R(\uparrow)$: $R + \frac{3}{5}T - 9.8m = 0 \Rightarrow R = 9.8m - \frac{3}{5}T$

$F = 9.8m\mu - \frac{3}{5}\mu T$

$R(\rightarrow)$: $\frac{4}{5}T - (9.8m\mu - \frac{3}{5}\mu T) = ma$

$\Rightarrow a = \frac{4}{5m}T - 9.8\mu + \frac{3}{5m}\mu T$

$a = (\frac{4}{5m} + \frac{3}{5m}\mu)T - 9.8\mu$ as req.

15 a)

$\sin\theta = \frac{5}{13}$ and $\cos\theta = \frac{12}{13}$

$R(\uparrow)$: $R - 10 \times \frac{5}{13} - 4 \times 9.8 = 0 \Rightarrow R = \frac{2798}{65}$

$F = 0.2 \times \frac{2798}{65} = \frac{2798}{325}$

$R(\rightarrow)$: $10 \times \frac{12}{13} - \frac{2798}{325} = 4a \Rightarrow a = 0.155$ (3 SF)

$a = 0.16$ (2 SF – AQA)

15 b)

$R(\uparrow)$: $R - 10 \times \frac{5}{13} - 4.1 \times 9.8 = 0 \Rightarrow R = \frac{28617}{650}$

$F = 0.2 \times \frac{28617}{650}$

$R(\rightarrow)$: $10 \times \frac{12}{13} - 0.2 \times \frac{28617}{650} = 4.1a \Rightarrow a = 0.104$ (3 SF)

$a = 0.10$ (2 SF – AQA)

15 c)

$R(\uparrow)$: $R - 10 \times \frac{5}{13} - 4.5 \times 9.8 = 0 \Rightarrow R = \frac{6233}{130}$

$F = 0.2 \times \frac{6233}{130} = \frac{6233}{650}$

$R(\rightarrow)$: $10 \times \frac{12}{13} - \frac{6233}{650} = -\frac{233}{650} \Rightarrow$ not moving

$\Rightarrow a = 0$

16 a)

$R(\rightarrow)$: $P - F = 0 \Rightarrow P = F$

$R(\rightarrow)$: $2P - F = 8 \times 0.25 \Rightarrow P = 2$

16 b)

$R(\uparrow)$: $R - 8 \times 9.8 = 0 \Rightarrow R = 78.4$

$F = 2 \Rightarrow 2 = 78.4\mu$

$\mu = \frac{5}{196} = 0.0255$ (3 SF)

$\mu = 0.026$ (2 SF – AQA)

16 c)

$R(\uparrow)$: $R - (8 + m) \times 9.8 = 0 \Rightarrow R = 78.4 + 9.8m$

$F = \frac{5}{196} \times (78.4 + 9.8m) = 2 + 0.25m$

$R(\rightarrow)$: $4 - (2 + 0.25m) \geq 0$

$2 - 0.25m \geq 0$

$2 \geq 0.25m \Rightarrow m \leq 8$

17 a)

$R(\uparrow)$: $R + 8\sin 30° - 1.2 \times 9.8 = 0 \Rightarrow R = 7.76$

$F = 0.2 \times 7.76 = 1.552$

$R(\rightarrow)$: $8\cos 30° - 1.552 = 1.2a$

$\Rightarrow a = 4.48$ ms^{-2} (3 SF)

$a = 4.5$ ms^{-2} (2 SF – AQA)

17 b)

$v = 0 + 4.480169359\ldots \times 5 = 22.4$

$R(\uparrow)$: $R - 1.2 \times 9.8 = 0 \Rightarrow R = 11.76$

$F = 0.2 \times 11.76 = 2.352$

$R(\rightarrow)$: $-2.352 = 1.2a \Rightarrow a = -1.96$ ms^{-2}

$0 = 22.4^2 - 2 \times 1.96s \Rightarrow s = 128$ m (3 SF)

$s = 130$ m (2 SF – AQA)

18 a)

$R(\uparrow)$: $R + 20\sin 30° - 5 \times 9.81 = 0 \Rightarrow R = 39.05$

$F = 0.4 \times 39.05 = 15.62$

$R(\rightarrow)$: $20\cos 30° - 15.62 = 5a$

$a = 2\sqrt{3} - 3.124 \Rightarrow a = 0.340$ ms^{-2} (3 SF)

18 b)

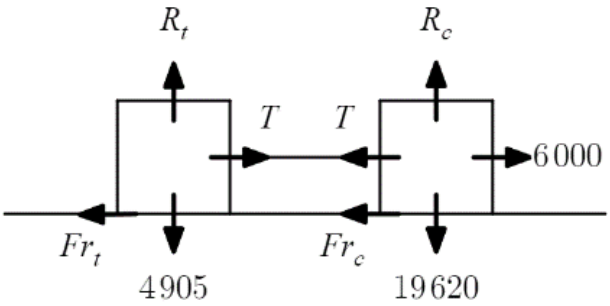
$R(\uparrow)$: $R - 5 \times 9.81 = 0 \Rightarrow R = 49.05$

$F = 0.4 \times 49.05 = 19.62$

$R(\rightarrow)$: $P - 19.62 = 5(2\sqrt{3} - 3.124)$

$\Rightarrow P = 21.3$ (3 SF)

19 a)



19 b)

On both (consider totals):

$R(\uparrow)$: $R - 2500 \times 9.81 = 0 \Rightarrow R = 24525$

$F = 0.1 \times 24525 = 2452.5$

$R(\rightarrow)$: $6000 - 2452.5 = 2500a$

$a = 1.419$ ms^{-2}

$a = 1.42$ ms^{-2} (3 SF – AQA)

19 c)

On Trailer:

$R(\uparrow)$: $R_T - 500 \times 9.81 = 0 \Rightarrow R_T = 4905$

$F_T = 0.1 \times 4905 = 490.5$

$R(\rightarrow)$: $T - 490.5 = 500 \times 1.419$

$T = 1200$ N

19 d)

The tow bar is modelled as a light inextensible string. There are no other resistances other than friction. The tow bar is horizontal.

20 a)

On both:

$R(\uparrow)$: $R - 4250 \times 9.81 = 0 \Rightarrow R = 41692.5$

$F = \mu \times 41692.5$

$R(\rightarrow)$: $10000 - \mu \times 41692.5 = 4250 \times 1.5$

$\mu = \frac{1450}{16677}$

$\mu = 0.0869$ (3 SF)

20 b)

On Trailer:

$R(\uparrow)$: $R - 750 \times 9.81 = 0 \Rightarrow R = 7357.5$

$F = \frac{1450}{16677} \times 7357.5 = \frac{10875}{17}$

$R(\rightarrow)$: $T - \frac{10875}{17} = 750 \times 1.5$

$T = 1764.705882\ldots$ N

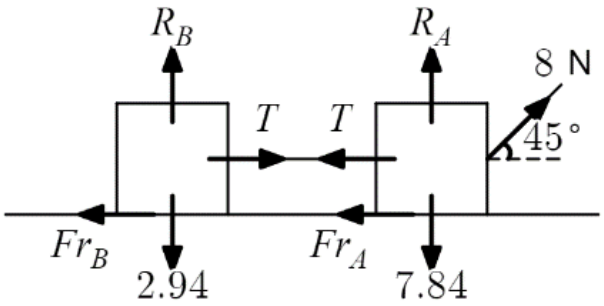
$T = 1770$ N (3 SF)

20 c)

Light: the mass of the towbar is negligible,

Inextensible: acceleration of the masses is the same.

21 a)



21 b)

On A: $R(\uparrow)$: $R_A - 7.84 + 8\sin 45 = 0$

$\Rightarrow R_A = 7.84 - 4\sqrt{2}$

$\Rightarrow Fr_A = 0.95(7.84 - 4\sqrt{2}) = 2.07$ N

$Fr_A = 2.1$ N (2 SF – AQA)

21 c)

On B: $R(\uparrow)$: $R_B - 2.94 = 0 \Rightarrow R_B = 2.94$

$\Rightarrow Fr_B = 0.95 \times 2.94 = 2.793$ N

$Fr_B = 2.8$ N (2 SF – AQA)

21 d)

On both: $R(\rightarrow)$:

$8\cos 45 - 0.95(7.84 - 4\sqrt{2}) - 2.793 = (0.8 + 0.3)a$

$a = 0.718$ ms^{-2}

$a = 0.72$ ms^{-2} (2 SF – AQA)

21 e)

On truck: $R(\rightarrow)$: $T - 2.793 = 0.3 \times 0.718$

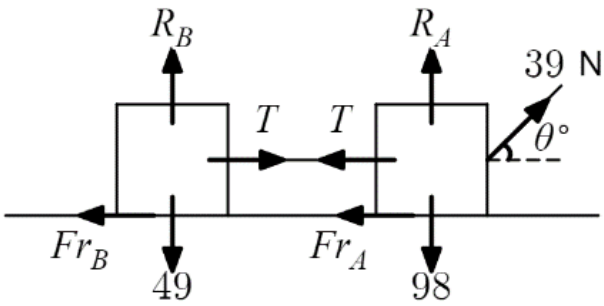
$T = 3.01$ N

$T = 3.0$ N (2 SF – AQA)

21 f)

Both strings are taut/rigid.

22 a)



22 b)

$\sin\theta = \frac{5}{13}, \cos\theta = \frac{12}{13}$

On B:

$R(\uparrow)$: $R_B - 49 = 0 \Rightarrow R_B = 49$

$Fr_B = 49\mu$

$R(\rightarrow)$: $T - 49\mu = 5 \times -3.5$

On A:

$R(\uparrow)$: $R_A + 39 \times \frac{5}{13} - 98 = 0 \Rightarrow R = 83$

$Fr_A = 83\mu$

$R(\rightarrow)$: $39 \times \frac{12}{13} - T - 83\mu = 10 \times -3.5$

Solving $T - 49\mu = -17.5$ and $-T - 83\mu = -71$

simultaneously gives $-132\mu = -88.5$

$\mu = \frac{59}{88} = 0.670$ (3 SF)

$\mu = 0.67$ (2 SF – AQA)

22 c)

$T = \frac{1351}{88} = 15.4$ N

$T = 15$ N (2 SF – AQA)

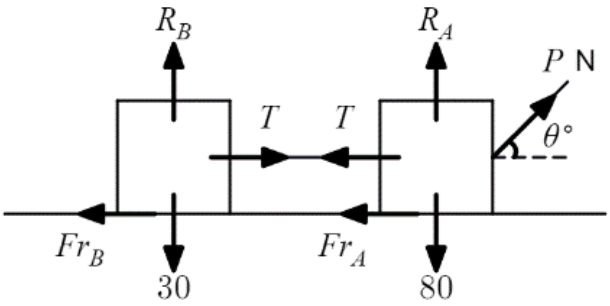
22 d)

The string connecting A and B is horizontal.

22 e)

$s = 0 - \frac{1}{2} \times -3.5 \times 10^2 = 175$ m

23 a)



23 b)

On B: $R(\uparrow)$: $R_A - 30 = 0 \Rightarrow R_A = 30$

$6 = 30\mu \Rightarrow \mu = 0.2$

23 c)

On B: $R(\rightarrow)$: $T - 6 = 3 \times 2.5 \Rightarrow T = 13.5$

23 d)

$\sin\theta = \frac{7}{25}, \cos\theta = \frac{24}{25}$

On A:

$R(\uparrow)$: $R_B + \frac{7}{25}P - 80 = 0 \Rightarrow R_B = 80 - \frac{7}{25}P$

$Fr_A = 0.2(80 - \frac{7}{25}P) = 16 - \frac{7}{125}P$

$R(\rightarrow)$: $\frac{24}{25}P - 13.5 - (16 - \frac{7}{125}P) = 8 \times 2.5$

$\frac{127}{125}P - 29.5 = 20$

$P = \frac{12375}{254} = 48.7$ N (3 SF)

$P = 50$ N (1 SF – AQA)

23 e)

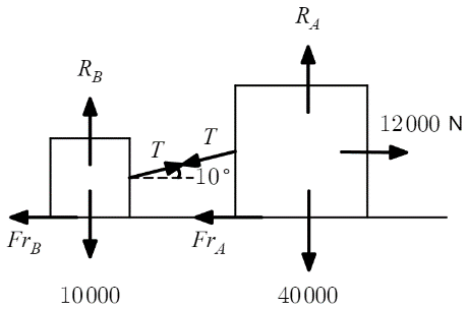
$Fr_A = 16 - \frac{7}{125} \times \frac{12375}{254} = 13.3$ N (3 SF)

$Fr_A = 10$ N (1 SF – AQA)

23 f)

$20 = 0 + \frac{1}{2} \times 2.5 \times t^2 \Rightarrow t = 4$ s

24 a)



24 b)

On Both:

$R(\uparrow)$: $R - 50000 = 0 \Rightarrow R = 50000$

$R(\rightarrow)$: $12000 - F = 5000 \times 0.5 \Rightarrow F = 9500$

$9500 = 50000\mu \Rightarrow \mu = 0.19$

24 c)

On B:

$R(\uparrow)$: $R - 10000 + T\sin 10 = 0$

$\Rightarrow R = 10000 - T\sin 10$

$\Rightarrow F = 0.19(10000 - T\sin 10)$

$= 1900 - 0.19T\sin 10$

$R(\rightarrow)$: $T\cos 10 - 1900 + 0.19T\sin 10 = 1000 \times 0.5$

$T(\cos 10 + 0.19\sin 10) = 2400$

$\Rightarrow T = 2358.025017\ldots$ so $T = 2360$ N (3 SF)

$T = 2000$ N (1 SF – AQA)

25 a)

On both:

$R(\uparrow)$: $R - 4m \times 10 = 0 \Rightarrow R = 40m$

$R(\rightarrow)$: $18m - F = 4m \times 2 \Rightarrow F = 10m$

$\mu = \frac{10m}{40m} = 0.25$ as req.

25 b)

On B:

$R(\uparrow)$: $R_B - 3m \times 10 + T\sin\theta = 0$

$\Rightarrow R_B = 30m - T\sin\theta$

$F_B = 7.5m - 0.25T\sin\theta$

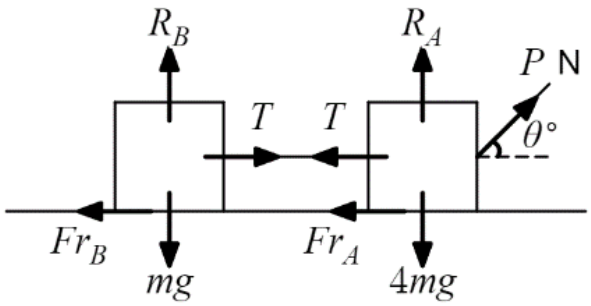
$R(\rightarrow)$: $T\cos\theta - 7.5m + 0.25T\sin\theta = 3m \times 2$

$T\cos\theta + 0.25T\sin\theta = 13.5m$

$T(\cos\theta + 0.25\sin\theta) = 13.5m$

$T = \frac{13.5m}{\cos\theta + 0.25\sin\theta} = \frac{54m}{4\cos\theta + \sin\theta} = \frac{54m}{\sin\theta + 4\cos\theta}$ as req.

26 a)



26 b)

$\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$

On both:

$R(\uparrow)$: $R + \frac{3}{5} \times 25 - 5mg = 0 \Rightarrow R = 5mg - 15$

$10 = \mu(5mg - 15)$

$\Rightarrow \mu = \frac{10}{5mg-15} = \frac{2}{mg-3}$ as req.

26 c)

On both:

$R(\rightarrow)$: $\frac{4}{5} \times 25 - 10 = 5ma$

$\Rightarrow a = \frac{10}{5m} = \frac{2}{m}$ ms^{-2} as req.

26 d)

On B:

$R(\uparrow)$: $R_B - mg = 0 \Rightarrow R_B = mg$

$R(\rightarrow)$: $T - (\frac{2}{mg-3})mg = m \times \frac{2}{m}$

$T = 2 + (\frac{2}{mg-3})mg = \frac{2(mg-3)}{mg-3} + \frac{2mg}{mg-3} = \frac{4mg-6}{mg-3}$